

Guilherme Jacob

Survey Sampling, SAE, Social and Economic Statistics

contact

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languages

Portuguese - native
English - fluent

softwares

R, Stan,
GIS, CPro,
SQL, LaTeX & Office

expertise

complex surveys,
small area estimation,
non-probability samples,
record linkage,
statistical modeling,
hierarchical models,
statistical programming,
poverty and inequality.

education

- | | |
|-----------|---|
| 2021–2025 | Ph.D. in Population, Territory, and Official Statistics ENCE/IBGE
National School of Statistical Sciences. Supervisor: Pedro Luis do Nascimento Silva, Ph.D.
Thesis defended in February 2025. CAPES scholarship.
Visiting Ph.D. student (sandwich program) at the University of Southampton, 2023–2024, supervised by Nikos Tzavidis, Ph.D. |
| 2019–2021 | M.Sc. in Population, Territory, and Official Statistics ENCE/IBGE
National School of Statistical Sciences. Supervisor: Pedro Luis do Nascimento Silva, Ph.D.
<i>Estimation of gross labor flows with the Brazilian Continuous Household Survey</i>
Dissertation defended in February 2021. CAPES scholarship. |
| 2020 | Sampling Program for Survey Statisticians Univ. of Michigan
Leslie Kish Fellow. 73rd Annual Summer Institute in Survey Research Techniques, Institute for Social Research.
Courses: Methods of Survey Sampling; Analysis of Complex Survey Data; Workshop in Survey Sampling Techniques. |
| 2014 | Refresher Course in Economic Theory UFRJ
Rio de Janeiro Federal University. <i>Refresher course in microeconomics, macroeconomics, mathematics, statistics, econometrics and Brazilian economy.</i> |
| 2009–2013 | B.Sc. in Economics UFAM
Amazonas Federal University.
<i>Brazilian Economic Model in the 2000s: an economic analysis from Keynes and Furtado</i>
My bachelor thesis proposed a new formula for the Mundell-Fleming model, including a hypothesis on the relationship between consumption, imports and inequality.
Overall Score: 8.39 out of 10.0. |
| 2008–2013 | B.A. in Law FMF
Faculdade Martha Falcão DeVry.
<i>Socioeconomic Etiology of Crime</i>
My bachelor thesis studied the relationship of poverty, inequality, and education as causative factors of criminality, combining the analysis of criminological theories and empirical testing of statistical models.
Overall Score: 7.98 out of 10.0. |

2025–present	Getulio Vargas Foundation (FGV-IBRE) <i>Associate Researcher / Consultant, Brazilian Institute of Economics</i> Analysis of survey data and planning of socioeconomic surveys for the Public Statistics Superintendency. Methodological reports on sample size determination and on inference for survey data, including non-probability samples.	Brazil
2022–2023	United Nations Population Fund (UNFPA) <i>Individual Consultant</i> Statistical data matching and estimation of rates and indicators of the Post-Enumeration Survey (PPE) of the 2022 Demographic Census. Support to the IBGE Coordination of Evaluation of Census 2022, in the evaluation of coverage and quality and in the monitoring of PPE results.	Brazil
2019	Brazilian Institute for Geography and Statistics (IBGE) <i>Consultant</i> Questionnaire evaluation and consistency checks for the National Health Survey (PNS).	Rio de Janeiro
2018–2019	United Nations Children's Fund (UNICEF) <i>Consultant</i> Analysis of school exclusion and overage students using administrative records and complex survey data.	Brasília, Federal District
2015–2017	United Nations Children's Fund (UNICEF) <i>UNV Monitoring and Evaluation Officer</i> Monitoring and evaluation of public policies, focusing on children and adolescents. Analysis of health, education, violence and poverty data from public records and complex survey samples. Also working on innovative initiatives using mobility data from mobile phones, information poverty measurement, and supporting Zika virus epidemiological data analysis.	Brasília, Federal District
2014–2015	Oswaldo Cruz Foundation (Fiocruz) / Ministry of Health <i>Consultant</i> Consultancy on data collection and analysis of social statistics for the Situation Analysis of children in the Brazilian Amazon, combining equity focus and the rights-based approach. This diagnostic aims to promote more inclusive policies for the most disadvantaged children.	Manaus, Amazonas
2014	United Nations Children's Fund (UNICEF) <i>Consultant</i> Collect data and support the analysis of the social and economic statistics of 50 municipalities with the worst living conditions in the Brazilian Amazon. Special focus on children and adolescents.	Manaus, Amazonas
2014	Amazonas Sustentável Foundation <i>Consultant</i> Questionnaire data analysis of Mathematics and Portuguese test performance of students in riverside schools of 8 municipalities of the Amazonas State.	Manaus, Amazonas

2014	Norte MC Consultores Ltda. <i>Consultant</i> Seasonal adjustment of industrial production by technological intensity data from January, 2002 to December, 2013. Development of programming routines for the X-12-ARIMA program. Results used in "Carta IEDI n.619", publication of the Institute of Studies for Industrial Development (IEDI).	Manaus, Amazonas
2013–2014	United Nations Children's Fund (UNICEF) <i>Consultant</i> Data collection and cluster analysis of municipalities in the Brazilian Amazon. Analysis of socioeconomic indicators. This task was part of UNICEF "Selo Município Aprovado" (Municipal Seal of Approval).	Manaus, Amazonas

academia

2023	Thesis Committee Member Federal University of Minas Gerais. Monograph for the Specialization in Statistics on the modeling of gross flows using the Brazilian Continuous Household Survey (PNAD Contínua), focused on informal labor market flows.	UFMG
2021–2025	Teaching Internship (undergraduate) EST052 – Sampling (supervisor: Maurício Lila, D.Sc.) and EST054 – Computational Statistics II (supervisor: Gustavo Rocha, D.Sc.).	ENCE/IBGE
2012–2013	Research Assistant Institute of Applied Economic Research. Research conducted in partnership between UFAM and IPEA (project: Regional Input-Output Matrix). Extraction and analysis of income and consumption micro data from the Family Budget Survey (2002-2003). Also, author of the methodological report for IPEA.	IPEA
2011–2012	Scientific Initiation Fellow (PIBIC) Questionnaire analysis, data collection and analysis regarding the profile of the Economics undergraduates in UFAM. Supervisor: Andreia Brasil Santos.	UFAM/CNPq
2010–2012	Teaching Assistant Fellow in Introduction to Economics: Macroeconomics (2010) and Microeconomic Theory I (2011); volunteer in International Economics (2012).	UFAM

technical and scientific production

2025	Article in Scientific Journal Quarterly small area estimates of extreme poverty in Brazil using transformed Fay-Herriot models Co-author, with N. Tzavidis, A. Luna-Hernandez and P. L. N. Silva. <i>Journal of Survey Statistics and Methodology</i> . Area-level model for quarterly extreme poverty rates in the geographic strata of the Brazilian Continuous Household Survey (PNAD Contínua). Developed during the sandwich Ph.D. at the University of Southampton.	
2025	Technical Reports (FGV-IBRE) Sample size and inference for socioeconomic surveys Author. Series of methodological reports on sample size determination for price index surveys and for the estimation of proportions, and on inference for survey data, including non-probability samples and homogeneity tests across independent surveys. Not public.	

2022–2023	Technical Reports (IBGE/UNFPA)	Evaluation of the 2022 Demographic Census
	Author. Series of reports on probabilistic record linkage, and on the estimation of coverage rates and content evaluation of Census 2022 based on the Post-Enumeration Survey (PPE). Not public.	
2021	IASS Cochran-Hansen Prize	Estimation of gross flows under nonresponse and complex sampling
	Paper awarded the Cochran-Hansen Prize by the International Association of Survey Statisticians. Co-authored with Pedro Silva.	
2021	Article in Scientific Journal	O Futuro do Pacote convey
	Co-author, with A. Damico. <i>Revista Brasileira de Estatística</i> , v. 79, pp. 64–74. Special issue in memory of Prof. Djalma Pessoa.	
2018	UNICEF Brazil Publication	Panorama da distorção idade-série no Brasil
	Statistical report about overage students in basic education.	
2017	Guest Reviewer in Scientific Journal	Journal of Official Statistics
	Special edition for the "New Techniques and Technologies for Statistics" Conference (NTTS), with the Collaboration in Research and Methodology for Official Statistics (CROS/EUROSTAT). Editorial Collaborators, 2017, 33(4), pp. 1093-1099.	
2017	Co-author of Book Chapter	Análise de dados utilizando as libraries survey e convey do R
	Chapter in "Statistics With R: Inovação no Mercado de Trabalho", a book composed of articles presented at the <i>International Seminar of Statistics with R</i> at the Fluminense Federal University.	
2017	UNICEF Brazil Publication	Caderno de Resultados Edição 2013–2016
	Team member (data analysis and report). Reduction of inequalities and advances in the rights of children and adolescents in municipalities of the Semi-arid and the Amazon.	
2017	UNICEF Brazil Publication	Boas Práticas: água e saneamento nas escolas do Semiárido
	Team member (data analysis and report). Report on good practices on WASH (water and sanitation) for schools in the Brazilian Semi-arid.	
2016	UNICEF Brazil Working Paper	Statistical inference for poverty and inequality measures using R "convey" package
	Author. Part of UNICEF Monitoring and Evaluation Team capacity building.	
2016	R package	convey: Income concentration analysis with complex survey samples
	Co-author. Focus on inequality measurement, subgroup decomposition and variance estimation algorithms. Used by IBGE in the production of official statistics.	

presentations and teaching

2024	Conference presentation	Small Area Estimation 2023-24 Conference
	Session "Survey Design and SAE for Measuring Household Economic Conditions": quarterly extreme poverty estimates with the Brazilian Continuous Household Survey.	
2022	Short course	XXII National Meeting of Population Studies (ABEP)
	"Measuring Inequality and Poverty with the PNAD Contínua".	
2022	Oral communication	24th National Symposium on Probability and Statistics (SINAPE)
	Quarterly poverty rates with the Brazilian Continuous Household Survey using multiple imputation.	

2021	Seminar	IBGE Seminar
	Estimation of labor market status flows with the Brazilian Continuous Household Survey.	
2020	Seminar	Dissertation Seminar, PPG in Population, Territories and Public Statistics
	Estimation of labor market status flows with the Brazilian Continuous Household Survey.	

training courses

2019	Survey Sampling Training Course	DHS eLearning
2017	Results-Based Management (RBM)	UNICEF Agora Platform
2017	United Nations Human Rights Responsibilities	UNICEF Agora Platform
2016	Immunization Coverage Data	UNICEF Agora Platform
2016	Introduction to Ethics in Evidence Generation	UNICEF Agora Platform
2016	Gender Equality UN Coherence and You	UNICEF Agora Platform
2016	The Data Scientist's Toolbox	Coursera
2015	Ethics and Integrity at UNICEF	UNICEF Agora Platform